

INLAND WATER'S CHALLENGES IN UNGAUGED INTERNATIONAL BASINS:
IDENTIFYING THE DESICCATION DRIVERS OF ENDORHEIC HAMUN LAKES

by

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DEDICATION

I dedicate my dissertation work to my spouse and parents. A special feeling of gratitude to my loving spouse, Zahra whose words of encouragement and support were a key in completion of this study. My parents, who have been my source of inspiration, guide, who continually provided their moral, emotional support.

I also dedicate this dissertation to all my friends who have given me inspiration every time.

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ABSTRACT

Inland waters face unprecedented pressures from climatic and anthropogenic stresses, causing their recession and desiccation globally. Climate change is increasingly blamed for such environmental degradation, but in many regions, direct anthropogenic pressures compound, and sometimes supersede, climatic factors. This study examined a human-environmental system – the terminal Hamun Lakes on the Iran-Afghanistan border – that embodies amplified challenges of inland waters. Satellite and climatic data from 1984-2019 were fused, which documented that the Hamun Lakes lost 89% of their surface area between 1999 and 2001 (3,809 km² versus 410 km²), coincident with a basin-wide, multi-year meteorological drought. The lakes continued to shrink afterwards and desiccated in 2012, despite the above-average precipitation in the upstream basin. Rapid growth in irrigated agricultural lands occurred in upstream Afghanistan in the recent decade, consuming water that otherwise would have fed the Hamun Lakes. Compounding upstream anthropogenic stressors, Iran began storing flood water that would have otherwise drained to the lakes, for urban and agricultural consumption in 2009. Results from a deep Learning model of Hamun Lakes' dynamics indicate that the average lakes' surface area from 2010-2019 would have been 2.5 times larger without *increasing* anthropogenic stresses across the basin. The Hamun Lakes' desiccation had major socio-environmental consequences, including loss of livelihood, out-migration, dust-storms, and loss of important species in the region.

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LIST OF ABBREVIATIONS

BSU	Boise State University
GC	Graduate College
TDC	Thesis and Dissertation Coordinator
AET	Actual EvapoTranspiration
a.k.a.	Also Known As
AChP	After Change Point
Afgh	Afghanistan
AGA	Average of Agricultural Area
ANDWI	Augmented Normalized Difference Water Index
ARMIA	AutoRegressive Integrated Moving Average
AWEI	Automated Water Extraction Index
AWEInsh	Automated Water Extraction Index - shadow
AWEIsh	Automated Water Extraction Index - non-shadow
BChP	Before Change Point
CDF	Cumulative Distribution Function
CE	Commission Error
CHIRPS	Climate Hazards Group InfraRed Precipitation with Station
CPC	Climate Prediction Center
CRU	Climatic Research Unit
DSM	Digital Surface Model

EPA	Environmental Protection Agency
ET	Ensemble of Trees
ETC	Extended Triple Collocation
ETM	Enhanced Thematic Mapper
EVI	Enhanced Vegetation Index
GEE	Google Earth Engine
GEV	Generalized Extreme Value
GI	Green Index
GPCC	Global Precipitation Climatology Centre
GR	Gaussian process Regressions
HRB	Helmand River Basin
HSV	Hue–Saturation–Value
km ²	Squared Kilometer
KS	Kolmogorov Smirnov
LSTM	Long Short-Term Memory
LSWI	Land Surface Water Index
MA	Moving Average
MAE	Mean Absolute Error
MCM	Million Cubic Meters
MK	Mann-Kendall
mm	Millimeter
MNDWI	Modified Normalized Difference Water Index
MODIS	Moderate Resolution Imaging Spectroradiometer

NBR	Normalized Burn Ratio index
NCEP/NCAR	National Centers for Environmental Prediction/National Center for Atmospheric Research
NDVI	Normalized Difference Vegetation Index
NDWI	Normalized Difference Water Index
NDWI	Normalized Difference Water Index
NIR	Near infrared
NOAA	National Oceanic and Atmospheric Administration
OE	Omission Error
OLI	Operational Land Imager
P	Precipitation
PERSIANN-CDR	Precipitation Estimation from Remotely Sensed Information using Artificial Neural Networks - Climate Data Record
PET	Potential EvapoTranspiration
PM	Particulate Matter
QA	Quality Assurance
RBF	Richards-Baker Flashiness Index
RGB	Red, Green, Blue
RMSE	Root Mean Squared Error
RT	Regression Trees
SARIMA	Seasonal AutoRegressive Integrated Moving Average
SAVI	Soil-Adjusted Vegetation Index
SB	Sub-Basin

SPI	Standardized Precipitation Index
SR	Surface Reflectance
SVM	Support Vector Machines
SWIR	shortwave infrared
TGI	Transpiration Greenness Index
TIRS	Thermal Infrared Sensor
TOA	Top of Atmosphere
TOA	Threshold
U.S.	United States
UCSB	University of California, Santa Barbara
WGI	Water-adjusted Green Index
WI	Water Index

CHAPTER ONE: INTRODUCTION

Lakes modulate critical hydrological, ecological and biogeochemical processes locally, regionally and globally (Pi et al., 2022). More than 100 million small to large lakes globally (Woolway et al., 2020) provide a variety of ecosystem services, such as providing habitat to a various species and supporting terrestrial and aquatic biodiversity (e.g., fish, amphibians, and local and migratory water birds), contributing to the local economy (e.g., tourism, fishing, and transportation), modulating local climate and weather (e.g., reducing maximum temperature), conserving soil (e.g., sedimentation in the lake bed and preventing sand/dust/salt storms), and in case of freshwater lakes offering freshwater for human and industrial consumption, among many other services (Woolway et al., 2020; Messenger et al., 2016). Various sources have estimated that lakes cover 2.67 to 5 million km² (1.8%-3.6% of the world's land surface area; Zhao et al., 2022; Messenger et al., 2016), but their surface area and volume are changing across the globe due to a variety of natural and anthropogenic forces (Woolway et al., 2020; Yao et al., 2023). While a warming climate has contributed to permafrost and glacier melting and an increase in receiving lakes area and storage volume (Grant et al., 2021), lakes in arid and semi-arid regions of the world are facing existential threats (Bai et al., 2011; Djoudi et al., 2013; Ashraf et al., 2019; Cooley et al., 2021). Indeed, more than half of the largest global lakes observed a statistically significant declining trend from 1992-2020 (Yao et al., 2023). A dramatic example of shrinking lakes in the arid regions is the Aral Sea, which lost 90% of its volume, shrunk 74% in surface area, and lost 23 m of its water